

Examiner
only

Option B – Medical Physics

7. (a) (i) Discuss how the properties of X-rays make them suitable for medical imaging. [3]

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(ii) An X-ray machine has an operating pd of 18 kV and a current of 12 mA. If only 0.5% of the power is converted into X-rays find:

I. the velocity with which the electrons strike the target; [2]

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II. the power of the emitted X-rays. [2]

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- (b) (i) The intensity, I , of an ultrasound beam decreases with the thickness, x , of a material according to the equation:

$$I = I_0 e^{-\mu x}$$

The half-value thickness, $x_{\frac{1}{2}}$ is the thickness of the material that reduces the intensity of an incident beam by 50 %. Show that:

$$\mu x_{\frac{1}{2}} = \ln 2 \qquad [2]$$

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- (ii) The half-value thickness of muscle for ultrasound of frequency 1.0 MHz is 2.7 cm. Determine the thickness of muscle required to reduce the intensity to 70 % of its original value. [3]

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(c) A patient has a suspected heart murmur possibly caused by a problem with their aortic valve. You have the choice of the following forms of medical imaging available:

MRI scan ultrasound B-scan fluoroscopy CT scan X-ray

Evaluate the effectiveness of each type of imaging in confirming the diagnosis. [5]

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- (d) (i) When discussing radiation exposure, medical scientists will mention the *absorbed dose*, D , and the *equivalent dose*, H . Explain the difference between these two terms. [2]

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- (ii) During treatment for a cancerous tumour using gamma radiation, a patient's lungs received an equivalent dose of 4 mSv. If the weighting factor of lung tissue is 0.12 calculate the effective dose. [1]

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